

VPA ICT — Grades 4–5 Term 3: Research & Coding

Term 3: Research & Coding — Weeks 25 – 36

Theme: Internet research, Scratch programming, digital citizenship | Duration: 40 minutes per lesson

Term Overview

Students develop critical research skills, learn responsible digital communication, and create interactive projects using Scratch. This term emphasizes information literacy, creative problem-solving, and ethical technology use.

Week-by-Week Scheme of Work

Week 25 — Internet Research Skills

Keywords: research 研究 (yánjiū), keyword 关键词 (guānjiàncí), search engine 搜索引擎 (sōusuǒ yǐnqíng), results 结果 (jiéguǒ), relevant 相关的 (xiāngguān de)

Activity 1 (15 min): "Keyword Challenge" — Given a broad question ("Tell me about animals"), students brainstorm specific keywords ("tiger habitat", "what do tigers eat"). Compare search results.

Activity 2 (15 min): "Search Comparison" — Same keyword in 3 search engines. Which gives the best results for kids?

Activity 3 (10 min): Note-taking practice — Read one article, write 3 facts in own words.

Formative: Uses specific keywords; finds relevant information.

Week 26 — Evaluating Sources

Keywords: reliable 可靠的 (kěkào de), trustworthy 值得信赖的 (zhídé xìnài de), author 作者 (zuòzhě), date 日期 (rìqī), fact 事实 (shìshí), opinion 观点 (guāndiǎn)

Activity 1 (15 min): "Real or Fake?" — Show 5 websites (some fake, some real). Students use checklist: Who wrote this? When? Does it look professional?

Activity 2 (15 min): "Spot the Difference" — Compare a Wikipedia article about sharks with a sensational blog post. Which is more reliable?

Activity 3 (10 min): Create a class "Trusted Sites" bookmark folder.

Formative: Can identify 2/3 reliability factors in a website.

Week 27 — Email Etiquette

Keywords: email 电子邮件 (diànzǐ yóujiàn), subject line 主题行 (zhǔtí háng), greeting 问候 (wènhòu), closing 结尾 (jiéwěi), CC 抄送 (chāosòng), attachment 附件 (fùjiàn)

Activity 1 (15 min): "Email Anatomy" — Label the parts of a professional email: subject, greeting, body, closing, signature.

Activity 2 (15 min): "Write a Professional Email" — Students write an email to a teacher requesting help with homework. Focus on polite language and clear subject.

Activity 3 (10 min): "Email Don'ts" — Show bad examples: ALL CAPS, no subject, slang. Fix them.

Formative: Email has all 5 parts and appropriate tone.

Week 28 — Digital Footprint

Keywords: digital footprint 数字足迹 (shùzì zújì), permanent 永久的 (yǒngjiǔ de), reputation 名誉 (míngyù), online identity 网络身份 (wǎngluò shēnfèn), privacy settings 隐私设置 (yǐnsī shèzhì)

Activity 1 (15 min): "What will Google say about me?" — Students draw their "ideal digital footprint" vs. "bad digital footprint."

Activity 2 (15 min): Discussion — "Can you delete something from the internet?" (No, not really!) Share examples.

Activity 3 (10 min): "Positive Posting" — Students write 3 positive, kind things they could post online.

Formative: Can explain what a digital footprint is and why it matters.

Week 29 — Introduction to Scratch

Keywords: Scratch Scratch (可视化编程工具), sprite 角色 (juésè), backdrop 背景 (bèijǐng), script 脚本 (jiǎoběn), block 积木块 (jīmù kuài), green flag 绿旗 (lǜqí)

Activity 1 (15 min): "Scratch Tour" — Interface exploration. Identify stage, sprite list, block palette, script area.

Activity 2 (15 min): "My First Script" — Make the cat say "Hello!" and move 10 steps when green flag clicked.

Activity 3 (10 min): Sprite customization — Change the cat's color, size, and costume.

Formative: Can run a program with event block and motion block.

Week 30 — Movement & Events

Keywords: coordinate 坐标 (zuòbiāo), x-axis X轴 (X zhóu), y-axis Y轴 (Y zhóu), glide 滑行 (huáxíng), point towards 指向 (zhǐxiàng)

Activity 1 (15 min): "Maze Game" — Students draw a simple maze backdrop. Program a sprite to navigate using arrow keys.

Activity 2 (15 min): "XY Treasure Hunt" — Sprite moves to specific coordinates to collect treasures.

Activity 3 (10 min): Debugging practice — A broken maze program. Students find and fix the bug.

Formative: Sprite moves in response to keyboard events.

Week 31 — Loops & Variables

Keywords: variable 变量 (biànlìang), score 分数 (fēnshù), repeat 重复 (chóngfù), forever 无限循环 (wúxiàn xúnhuán), repeat until 重复直到 (chóngfù zhídào)

Activity 1 (15 min): "Score-Keeping Game" — Create a variable called "Score." Increase it when sprite touches an object.

Activity 2 (15 min): "Repeat Patterns" — Use loops to make a sprite draw a square, then a triangle, then a star.

Activity 3 (10 min): "Which loop?" — Given scenarios, choose: repeat, forever, or repeat until.

Formative: Creates and uses a variable; chooses appropriate loop type.

Week 32 — If/Then & Sensing

Keywords: if/then 如果/那么 (rúguǒ/nàme), sensing 侦测 (zhēncè), touching 碰到 (pèngdào), key pressed 按下按键 (ànxià ànjiàn), ask/answer 询问/回答 (xúnwèn/huídá)

Activity 1 (15 min): "Quiz Game with Scoring" — Ask question → check answer → if correct, add 10 points; if wrong, say "Try again!"

Activity 2 (15 min): "Boundary Detection" — Sprite bounces when touching edge. Add sound effect.

Activity 3 (10 min): Challenge — Add a 3-question quiz with cumulative scoring.

Formative: Program uses if/then and sensing blocks correctly.

Week 33 — Sprites & Costumes

Keywords: costume 造型 (zàoxíng), animation 动画 (dòng huà), switch costume 切换造型 (qiē huàn zàoxíng), next costume 下一个造型 (xià yí ge zàoxíng)

Activity 1 (15 min): "Animated Story" — Create a walking animation by switching between 2 costumes.

Activity 2 (15 min): "Character Conversation" — 2 sprites with multiple costumes have a dialogue using "say" blocks.

Activity 3 (10 min): Draw your own costume in Scratch's paint editor.

Formative: Uses costumes to create animation or dialogue.

Week 34 — Sounds & Broadcasting

Keywords: broadcast 广播 (guǎng bō), message 消息 (xiāo xī), when I receive 当接收到 (dāng jiē shōu dào), sound 声音 (shēng yīn), play sound 播放声音 (bō fàng shēng yīn)

Activity 1 (15 min): "Interactive Story" — Sprite 1 says "Once upon a time..." then broadcasts "next scene." Sprite 2 receives message and continues story.

Activity 2 (15 min): Sound effects — Add appropriate sounds to story events (door opening, bird chirping).

Activity 3 (10 min): Peer testing — Can your partner follow the story without help?

Formative: Uses broadcast to coordinate multiple sprites.

Week 35 — Review & Debug

Activity 1 (20 min): "Bug-Fixing Challenge" — 5 broken Scratch projects. Students identify and fix bugs.

Activity 2 (15 min): Project polish time — Add finishing touches: title screen, instructions, credits.

Activity 3 (5 min): Self-assessment rubric completion.

Week 36 — End of Year Assessment

Project Assessment (30 min):

Students submit a complete Scratch project with:

Title screen with instructions

Minimum 2 sprites with movement

At least 1 variable (score, lives, timer)

Use of loops and if/then statements

Sound or broadcast feature

Research Report (10 min): Short paragraph answering: "How did you test and improve your project?"

Standards Mapping

Weeks	Standard	Description
25 - 26	CCSS.ELA.W.4.7	Conduct short research projects using several sources

Weeks	Standard	Description
27 – 28	ISTE 2.1	Cultivate and manage digital identity and reputation
29 – 35	CSTA 1B-AP-10	Create programs that use variables and events
29 – 35	CSTA 1B-AP-11	Decompose problems into smaller, manageable tasks